

Runna

Software Requirement Specification

Wanikkasit Nopthiraitthikun 652115043

Thanachai Naksomboon 662115020

Bachelor of Science

Software Engineering Program

College of Arts, Media, and Technology

Chiang Mai University

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Project Advisor

[Assistant Professor Dr. Noppon Choosri]

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Chapter 1 | Introduction

1.1 Purposes

The primary purpose of this Software Requirement Specification (SRS) is to provide a comprehensive and detailed description of all requirements for the **Runna** platform. This document serves as the foundational agreement between the development team and the project advisor, ensuring a shared understanding of the project's scope, features, and constraints.

1.2 Scope

The software requirement specification document describes the Runna system's requirements, including use case diagram and description, user requirements specification, and system requirement specification. The Runna system consists of features, as shown in the table below:

	Pct of task complete (%)	
	Progress I	Progress II
Feature 1: Authentication & Roles	100 %	100 %
2.1 Register	✓	✓
2.2 Login	✓	✓
2.3 Logout	✓	✓
2.4 Password Reset	✓	✓
2.5 Role-Based Access Control	✓	✓
Feature 2: Route Creation & Tracking	50%	100 %

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○ View Map ○ Custom Route Creation ○ Route Editing(Add/Remove Waypoints) ○ Statistic Calculation ○ Real-time Tracking (Start/Stop Run) ○ Run History Storage	✓ ✓ ✓ ✗ ✗ ✗	✓ ✓ ✓ ✓ ✓ ✓
Feature 3: Collaborative Pin System	60%	100 %
• Create Pin • View Pin on Map • Pin Categories • User Confirmation • Pin Status Management(Active/Expired)	✓ ✓ ✓ ✗ ✗	✓ ✓ ✓ ✓ ✓
Feature 4: AI Run Summary	65%	100%
4.1 Send Data to Gemini API 4.2 AI Analysis & Insight Generation 4.3 Recommendation Generation	✓ ✓ ✗	✓ ✓ ✓
Feature 5: Admin Dashboard	35%	100%
5.1 Manage users 5.2 Remove incorrect hazard reports 5.3 Manage route	✓ ✗ ✗	✓ ✓ ✓

1.3 User Characteristics

The system interacts with three main classes of users, each possessing distinct access rights and operational goals:

User Type	Definition				
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Guest	An unauthenticated user who can download the mobile application, view promotional screens, and register an account.
Runner (Authenticated Member)	A registered athlete who actively utilizes the core application loop. They plot manual routes, record runs via GPS, view historical analytical reports, and report/view route hazards.
Administrator	A backend supervisor responsible for data integrity, managing platform statistics, and reviewing or overriding flagged crowdsourced safety hazard pins.

1.4 Acronyms and Definitions

1.4.1 Acronyms

SRS: Software Requirement Specification

URS: User Requirement Specification

UC: Use Case

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AD: Activity Diagram

UD: Use case Description

API: Application Programming
Interface

CRUD: Create, Read, Update, Delete

JSON: JavaScript Object Notation

JWT: JSON Web Token

UUID: Universally Unique Identifier

TDE: Transparent Data Encryption

UI/UX: User Interface / User Experience

OSM: OpenStreetMap

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1.4.2 Definitions

Terms	Definition
Manual Route	A user-defined running path established by clicking or tapping sequential geo-coordinates on the map interface.
Run Session	An active tracking period during which the mobile client samples and caches the runner's real-time coordinates before transmitting them to the backend.
Hazard Pin	A crowdsourced map marker deployed by a runner to denote dynamic physical hazards (e.g., construction, aggressive animals, poor lighting).
AI Insights	Summary assessments generated dynamically at the end of a run session using LLMs (Gemini) to evaluate runner performance relative to their historical thresholds.

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Chapter 2 | Specification Requirement

2.1 Feature Description

Feature #1: Authentication & Profile Management

Description: Users can register, log in, and manage their personal runner profiles securely.

Actor: Guest / Runner

Details:

- **Registration:** Guest registers with full name, email, and password. System checks for unique emails and hashes passwords.
- **Login/Logout:** Runner logs in to get a secure session token and logs out to clear it.

Feature #2: Route Planning & Map Interaction

Description: Runners interact with map layers to plan custom routes and calculate distances.

Actor: member

Details:

- Manual Mapping: Traces lines along roads or trails by tapping map points based on OpenStreetMap.
- Route Storage: Saves custom route paths into the system for future active tracking sessions.

Feature #3: Active Run Tracking

Description: Records spatial coordinates during outdoor runs and syncs data to the backend in chunks.

Actor: member

Details:

- **Batch Uploads:** Sends localized batches of GPS point arrays to the server

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Feature #4: Post-Run Analytics

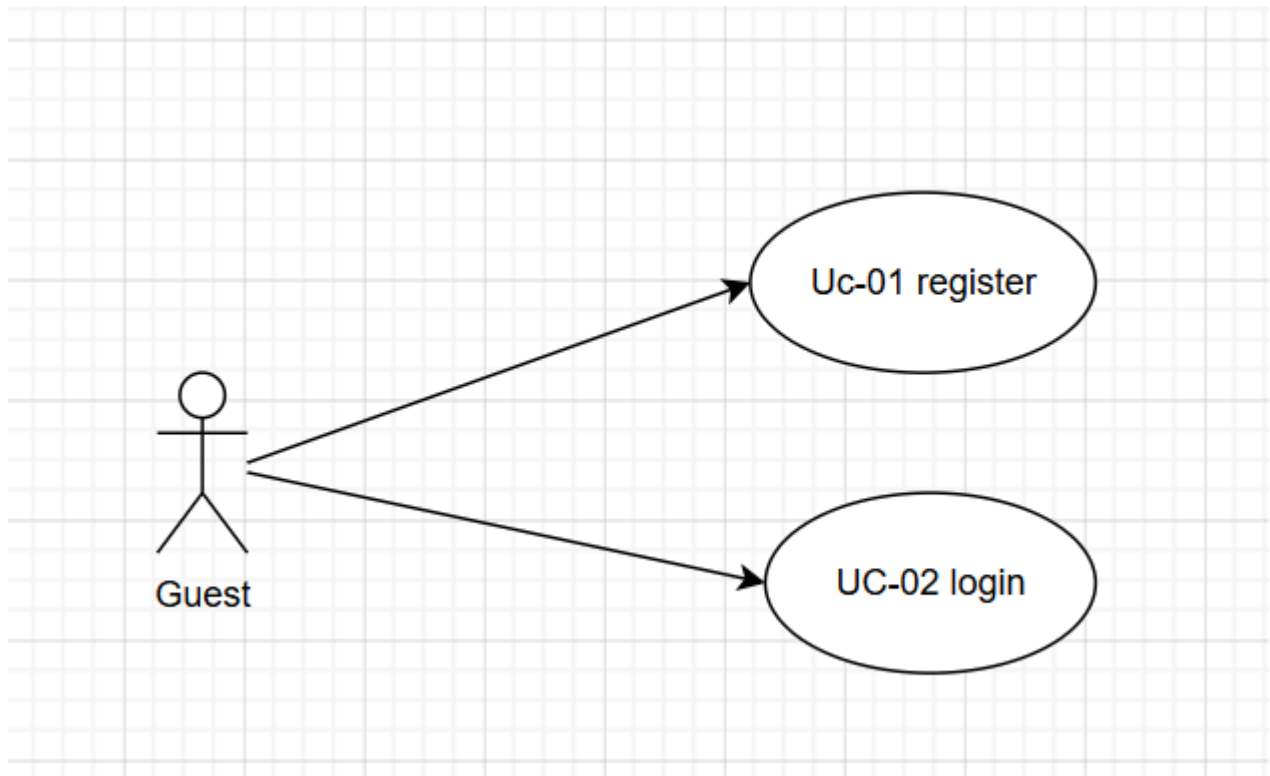
- **Description:** Processes raw spatial data to generate summaries, pace charts, and tailored feedback text.
- **Actor:** Runner
- **Details:**
 - **Mathematical Analytics:** Computes total active distance, total time, split paces, and average moving velocity.
 - **AI Insights Engagement:** Leverages the Gemini API to review personal metric history and generate supportive coaching text content.

Feature #5: Hazard Pin

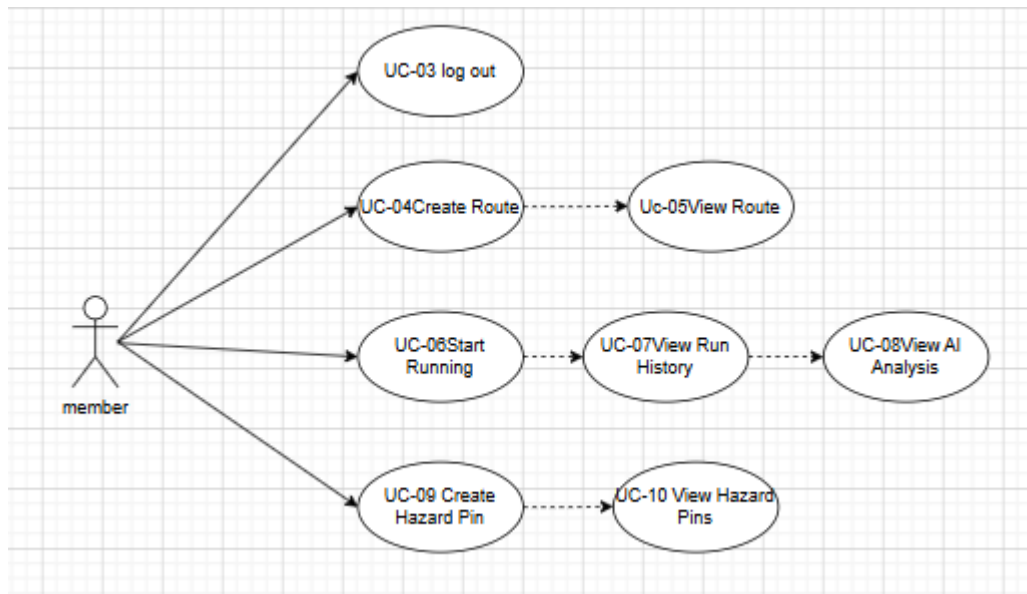
- **Description:** Runners post live localized hazards on maps while admins handle moderation actions.
- **Actor:** Runner / Admin
- **Details:**
 - **Hazard Pin Reporting:** Runners pin problems (e.g., dynamic construction, street flooding) onto current maps with brief notes.
 - **Administrative Overrides:** Admins use utility tools to adjust, close, or archive dynamic pins across production databases.

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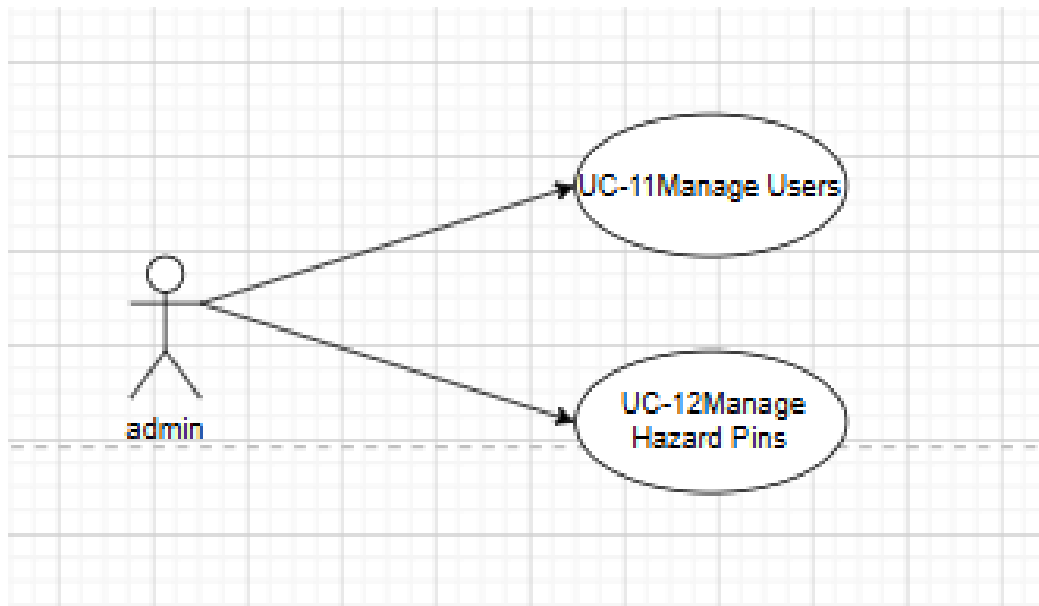
2.2 Use Case Diagram



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2.3 Use Case Description

UC-01: Guest registers an account

Use Case ID:	UD-01			
Use Case Name	Guest registers an account			
Created by	WN	Last update by		WN
Date created	16/06/2026	Last revision date		19/06/2026
Actor	Guests			
Description:	This use case allows an unregistered guest to create a new runner profile on the Runna platform by providing their full name, email, and password			
Trigger:	The guest selects the "Register" option on the sign-up page.			
Precondition:	The guest is on the Runna app interface and is not logged in			
Use Case Input Specification				
Input	Type	Description	Constraint	Example
Full Name	String	The user's full name for their runner profile.	Cannot be empty. Must contain only letters and spaces.	Jamal Johnson
Email Address	String	The user's unique email address which acts as the login ID	- Must follow standard email format. Must be unique in the database.	Test@example.com

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Password	String	A secure password for account authentication	- Must be at least 8 characters long with numbers and mixed cases.	123456Ab@
Confirm Password	String	A re-entry of the password to verify accuracy.	Must be an exact match to the Password field.	123456Ab@
Postcondition:	- A new account is successfully committed to the database, and the user is redirected to the login interface.			
Normal Flow:	Guests	System		
	1. Guest selects the "Register" option. 3. Guest enters all required information and submits the form.	2. Displays the registration form with all required fields.		

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		4. Receives registration data package from the client 5. Validates email format data [A1: Invalid email format] 6. Verifies that the email address is unique in the system [A2: email exists in the database] 7. Validates password confirmation matching. . [A3: Password not match] 8. Creates a new user account entry, securely hashes the password, saves information into PostgreSQL, and routes user to login view
Alternative Flow:	[A1: Invalid email format] A2.1: System displays: "Please enter email in the correct format." A2.2: Return to step 3 on Normal Flow. [A2: Email exists in the database] A2.1: Show "This email is already used." A2.2: Go to step 3 on Normal flow [A3: Password not match] A3.1: Show "The password does not match" A3.2: Go to step 3 on Normal flow	
Exception Flow:	None	
Assumption:	The guest has a working, stable internet connection	

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UC-02: Guest logs into the system

Use Case ID:	UD-02			
Use Case Name	Guest logs into the system			
Created by	WN	Last update by		WN
Date created	16/06/2026	Last revision date		19/06/2026
Actor	Guests			
Description:	Verifies existing account credentials to grant full feature access to the app's protected views.			
Trigger:	Guest enters their credentials and taps the primary "Login" interface option			
Precondition:	The user profile is already registered in the application's database.			
Use Case Input Specification				
Input	Type	Description	Constraint	Example
Full Name	String	The user's full name for their runner profile.	Cannot be empty. Must contain only letters and spaces.	Jamal Johnson
Email Address	String	The user's unique email address which acts as the login ID	- Must follow standard email format. Must be unique in the database.	Test@example.com

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Password	String	A secure password for account authentication	- Must be at least 8 characters long with numbers and mixed cases.	123456Ab@
Confirm Password	String	A re-entry of the password to verify accuracy.	Must be an exact match to the Password field.	123456Ab@
Postcondition:	- The system generates a secure JWT token, stores it in the app's local memory, and loads the Member Dashboard map view.			
Normal Flow:	Guests	System		
	1. Guest fills out the Email and Password fields on the login page, then clicks "Login".	2. System receives the authentication request package. 3. System checks if the provided email address exists in the database. [A1: Account Not Found]		

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		4. System verifies the password against the stored bcrypt hash. [A2: Wrong Password Credential] 5. System creates a secure session token payload, appends the appropriate authorization scope roles, and logs the user into the Member view.
Alternative Flow:	[A1: Account Not Found] -> System displays: "No account linked to this email address." -> Refresh login screen. [A2: Wrong Password Credential] -> System displays: "Incorrect password provided." -> Clear password input box.	
Exception Flow:	None	
Assumption:	-	

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UC-03: Member logs out of the system

Use Case ID:	UD-03			
Use Case Name	Member logs out of the system			
Created by	WN	Last update by		WN
Date created	16/06/2026	Last revision date		19/06/2026
Actor	Guests			
Description:	Terminates the active authenticated user session and securely clears local authorization parameters			
Trigger:	Member selects the "Log out" button inside their profile settings layout.			
Precondition:	The runner session state must be actively authenticated via a valid token parameter.			
Use Case Input Specification				
Input	Type	Description	Constraint	Example
-	-	-	-	-

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Postcondition:	- User logout	
Normal Flow:	member	System
	1. Member taps the "Log out" button component.	2. System detects the request, calls the token destruction routine, and clears the client's local storage keys. 3. System safely destroys the user context and routes the terminal view back to the Guest landing view.

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UC-04: Create Route

Use Case ID:	UD-04			
Use Case Name	Create Route			
Created by	WN	Last update by		WN
Date created	16/06/2026	Last revision date		19/06/2026
Actor	Member			
Description:	Allows a runner to plot out a custom run path by tapping sequence point nodes across map interface layers.			
Trigger:	Member selects the "Create Route" tool widget on the mapping screen.			
Precondition:	Member must be authenticated and the map widget components must be loaded.			
Use Case Input Specification				
Input	Type	Description	Constraint	Example
Route Title	String	-	-	Morning run

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Postcondition:	- The new route polyline path is linked to the runner profile and saved into the database, instantly unlocking the View Route capability.	
Normal Flow:	member	System
	1. Member clicks on the "Create Route" option. 3. Member taps points on roads or trails to trace their intended route line.	2. System initializes a clean canvas overlay on the map view. 4. System calculates distance dynamically based on the dropped points. 5. System saves the path polyline data array into SQL.

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UC-05: View Route

Use Case ID:	UD-05		
Use Case Name	View Route t		
Created by	WN	Last update by	WN
Date created	16/06/2026	Last revision date	19/06/2026
Actor	Member		
Description:	Fetches and displays previously planned custom routes onto the map view for activity selection.		
Trigger:	Member selects a specific path item inside their saved route menu list.		
Precondition:	At least one custom route record must be linked to the active member profile database entry.		

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Normal Flow:	Guests	System (CoachCall)
	1. Member opens their personal route storage panel. 3. Member selects a target route title to view.	2. System fetches all associated route polyline names and summaries from the server database. 4. System renders the matching polyline trajectory lines on the map.

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Alternative Flow:	-
Exception Flow:	None
Assumption:	Members can see the path and use it for running.

UC-06: Start Running

Use Case ID:	UD-06		
Use Case Name	Start Running		
Created by	WN	Last update by	WN
Date created	16/06/2026	Last revision date	19/06/2026
Actor	Member		
Description:	Launches the app's tracking engine to record background location changes and metrics while running outdoors		
Trigger:	Member presses the green "Start Run" tracking button widget		
Precondition:	App must be granted hardware device location access permissions (GPS).		

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Use Case Input Specification				
Input	Type	Description	Constraint	Example
-	-	-	-	-

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Postcondition:	- Creates an active track run entry, which unlocks the View Run History pipeline once finished.	
Normal Flow:	member	System
	1. Member taps the "Start Running" button 4. Member completes their run and holds down the "Stop" button interface component.	2. System requests GPS coordinate feeds from the mobile device hardware. 3. System records coordinates and active run metrics in the background. 5. System gathers the recorded coordinate path array and uploads it to the backend server.

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UC-07: View Run History

Use Case ID:	UD-07			
Use Case Name	View Run History			
Created by	WN	Last update by		WN
Date created	16/06/2026	Last revision date		19/06/2026
Actor	Member			
Description:	Displays an itemized list of all past running sessions completed by the runner profile.			
Trigger:	Member visits the "Activities History" tab panel view.			
Precondition:	The runner profile must contain previously completed and saved activity logs.			
Use Case Input Specification				
Input	Type	Description	Constraint	Example
-	-	-	-	-

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Postcondition:	- History shows	
Normal Flow:	Guests	System
	1. Member clicks on the "View Run History" option. 3. Guest enters all required information and submits the form.	2. System executes an indexed database call for the member's activity log IDs. 3. System displays the history list showing the duration, pace, and date for each past run. .

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Alternative Flow:	-
Exception Flow:	None
Assumption:	-

UC-08: View AI Analysis

Use Case ID:	UD-08		
Use Case Name	View AI Analysis		
Created by	WN	Last update by	WN
Date created	16/06/2026	Last revision date	19/06/2026
Actor	Member		
Description:	show tailored coaching advice text based on recent run tracking metrics.		
Trigger:	Member opens a completed activity log item to read performance feedback		
Precondition:	the target run activity entry must have successfully uploaded its metric data payload to the system backend .		

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Use Case Input Specification				
Input	Type	Description	Constraint	Example
-	-	-	-	-

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Postcondition:	- Insight, reasoning and recommend shows	
Normal Flow:	Guests	System
	1. Member clicks on the "View Run History" option.	2. System packages recent pacing data, distance arrays, and historical training loads into a secure text payload context. 3. System passes the payload to the backend Gemini API wrapper. 4. Gemini returns personalized, conversational coaching feedback tips. 5. System formats and presents the coaching text alongside performance texts on the user's screen. .

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Alternative Flow:	-
Exception Flow:	None
Assumption:	-

UC-09: Create Hazard Pin

Use Case ID:	UD-09		
Use Case Name	Create Hazard Pin		
Created by	WN	Last update by	WN
Date created	16/06/2026	Last revision date	19/06/2026
Actor	Member		
Description:	Lets members drop warning pins onto current maps to flag dynamic route hazards for the running community		
Trigger:	Member taps the "Report Hazard" floating icon widget on the active map display.		
Precondition:	the target run activity entry must have successfully uploaded its metric data payload to the system backend .		

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Use Case Input Specification				
Input	Type	Description	Constraint	Example
Hazard Type category selection	dropdown	Description notes	-	-

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Postcondition:	- Adds a new warning marker to the area maps, feeding directly into the system's shared View Hazard Pins map layer.	
Normal Flow:	Guests	System
	1. Member selects the "Create Hazard Pin" option. 3.Member selects a hazard category (e.g., Construction, Bad Lighting) and adds a brief description note 4.Member clicks "Publish Hazard".	2. System detects the runner's current map coordinates. 5.System saves the pin coordinate parameters to the global hazard table database layer. .

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Alternative Flow:	-
Exception Flow:	None
Assumption:	-

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UC-10: View Hazard Pins

Use Case ID:	UD-10			
Use Case Name	View AI Analysis			
Created by	WN	Last update by		WN
Date created	16/06/2026	Last revision date		19/06/2026
Actor	Member			
Description:	Downloads and renders active hazard markers posted by the community onto the current map view.			
Trigger:	Member pans across the map view or starts a route exploration session			
Precondition:	Active hazard records must exist within the runner's current geographic map view bounding box.			
Use Case Input Specification				
Input	Type	Description	Constraint	Example
-	-	-	-	-

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Postcondition:	- Show current pin on the map	
Normal Flow:	Guests	System
	1. Member opens the primary run exploration map view.	2. System reads the current map view area limits. 3..System fetches active hazard pin records inside those boundaries from the database. 4.System draws matching hazard alert icons on the map layer.

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Alternative Flow:	-
Exception Flow:	None
Assumption:	-

UC-11: Manage Users (admin)

Use Case ID:	UD-11		
Use Case Name	Manage Users		
Created by	WN	Last update by	WN
Date created	16/06/2026	Last revision date	19/06/2026
Actor	Admin		
Description:	check user states, and maintain data security.		
Trigger:	Admin opens the "User Management" portal tab inside the control console dashboard.		
Precondition:	Admin must authenticate through a secure, designated administrative login		

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endpoint channel.				
Use Case Input Specification				
Input	Type	Description	Constraint	Example
-	-	-	-	-

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Postcondition:	- Show admin dashboard	
Normal Flow:	Guests	System
	1. Admin opens the "Manage Users" console section. 3.Admin can browse the list, search for specific profiles, or adjust user access permissions to clean up spam accounts	2. System fetches user account records, showing registration histories and account statuses.

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Alternative Flow:	-
Exception Flow:	None
Assumption:	-

UC-12: Manage Hazard Pins (admin)

Use Case ID:	UD-12		
Use Case Name	Manage Hazard Pins		
Created by	WN	Last update by	WN
Date created	16/06/2026	Last revision date	19/06/2026
Actor	Admin		
Description:	Gives administrators moderation tools to review, update, or clear stale crowdsourced map safety flags.		
Trigger:	Admin opens the "Hazard Review" section inside the admin control panel.		
Precondition:	Crowdsourced pin records must be present in the centralized database tables.		

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Use Case Input Specification				
Input	Type	Description	Constraint	Example
-	-	-	-	-

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Postcondition:	- Show admin dashboard	
Normal Flow:	Guests	System
	1. Admin clicks on the "Manage Hazard Pins" option 3.Admin reviews reported descriptions or checks flag timestamps. 4.Admin can manually update pin details, mark hazards as resolved, or clear old flags from the database tables.	2. System returns an itemized summary list of all active map pins currently flagged across production maps.

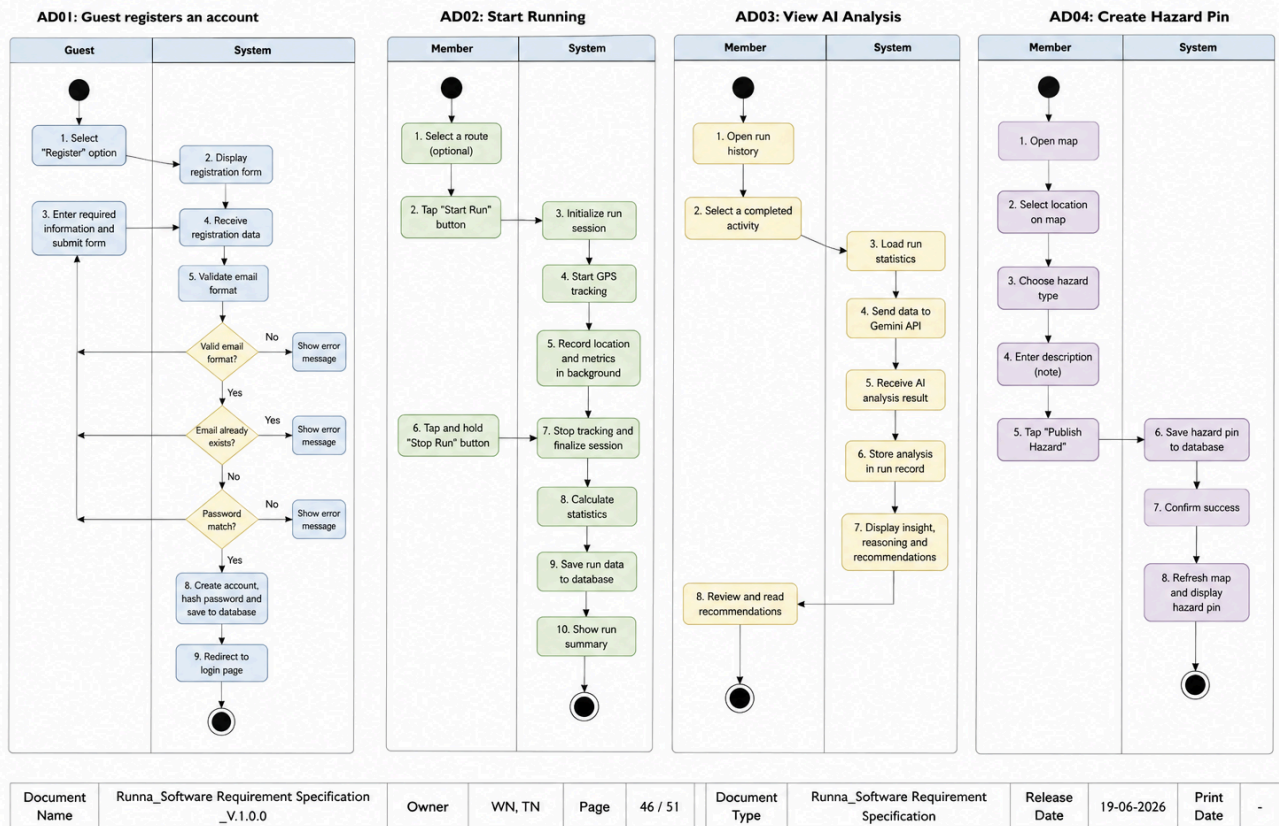
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Alternative Flow:	-
Exception Flow:	None
Assumption:	-

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2.4 Activity Diagram

2.4 Activity Diagram



2.5 User Requirement Specification

User Requirement Specification (URS)

- URS-01: System must allow guests to fill out registration parameters to onboard as members.
- URS-02: System must validate registered login parameters before issuing session permissions.
- URS-03: System must give authenticated users clear logout controls to safely clear local session states.
- URS-04: Members must be able to trace customized polyline tracks across interactive

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maps to build running paths.

- URS-05: Members must be able to load, view, and select their saved custom paths from an organized route menu list.
- URS-06: Members must be able to initiate tracking sessions that actively record device location logs in the background.
- URS-07: Members must be able to access historical activity logs to review past performances.
- URS-08: Members must be able to view automated AI coaching advice panels to help evaluate their running habits.
- URS-09: Members must be able to drop warning pins on specific map locations to flag active trail dangers.
- URS-10: Members must be able to see community-sourced hazard icons drawn clearly over their active map layers.
- URS-11: Administrators must have access to dedicated user overview consoles to inspect account statuses.
- URS-12: Administrators must be able to edit, archive, or remove hazard pins from global maps to ensure data accuracy.

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2.6 System Requirement Specification

- **SRS-01:** The system shall provide input forms with string length validation for full name, email, password, and confirm password fields.
- **SRS-02:** The system backend shall run regular expression checks to confirm incoming email fields follow standard legal web formats.
- **SRS-03:** The database layer shall use indexing constraints on user email tables to enforce global account uniqueness.
- **SRS-04:** The application interface shall require that password strings match exactly before unlocking submission buttons.
- **SRS-05:** The backend API shall pass password payloads through the **encrypt** algorithm, applying a salt factor of 10 before saving to PostgreSQL.
- **SRS-06:** The system shall issue cryptographically signed **JWT (JSON Web Tokens)** containing expiration fields set to 24 hours upon successful authentication.
- **SRS-07:** The FastAPI backend infrastructure shall include Cross-Origin Resource Sharing (CORS) configurations that explicitly permit headers coming from the client application.
- **SRS-08:** The mapping components shall capture and serialize geographic position nodes into standard **GeoJSON LineString** shapes for route persistence.
- **SRS-09:** The database shall connect running tracks with coordinate tracking nodes via one-to-many relationship rows in PostgreSQL tables.
- **SRS-10:** The system tracking loop shall accept chunk data packets (**POST /api/runs/{run_id}/points**) to continuously save running logs.
- **SRS-11:** The system shall call external Gemini API endpoints to process activity matrices and generate customized coaching text blocks.

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- **SRS-12:** If external AI services encounter an outage, the platform shall process running telemetry through a fallback mathematical statistics module to keep the metrics dashboard accurate.
- **SRS-13:** The map layer shall use spatial queries to fetch and draw community hazard pins located within the user's view bounding box.
- **SRS-14:** The administrative console dashboard shall provide dedicated **PUT** and **DELETE** endpoints to modify or clear target hazard table rows.

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